Claude CLI for Research: Cheat Sheet

AI-Empowered Research · CERP³ · Washington University in St. Louis · 2026-10-16

Commands come from the workshop setup guides (`part2_setup/setup_guides/` at github.com/YungChun0303/ai_workshop_edu), verified 2026-10-12.

1. Install and Verify

	Command
macOS (Terminal)	`curl -fsSL https://claude.ai/install.sh | bash`
Windows (PowerShell)	`irm https://claude.ai/install.ps1 | iex`
Windows (Command Prompt)	`curl -fsSL https://claude.ai/install.cmd -o install.cmd && install.cmd && del install.cmd`
Verify, then sign in	`claude --version` · `claude doctor` (read-only check) · `claude` (opens your browser)

- **Windows:** `PS C:\...>` is PowerShell and `C:\...>` is Command Prompt. `'irm' is not recognized` means you're in Command Prompt. You don't need Administrator rights or WSL.
- **Not recognized** (`claude` or `Rscript`)? Open a new terminal. On Windows, add R's `bin` folder to your PATH (guide §2, Step 5). **Never delete `~/.local/bin/claude`**: that file is Claude Code.

2. WashU Access

- **Claude Edu** includes Claude Code. It is approved for FERPA data and institutional IP, **not PHI**.
- **Claude Secure** is approved for PHI and FERPA data, but has **no Claude Code**.
- **Sign in:** claude.ai → WashU email → *Continue with SSO*. **Students:** confirm your seat includes Claude Code (it.washu.edu/claude-for-education).

3. Connect R (optional)

1. In R, once: `install.packages("mcptools")`
2. In the terminal, once:

```
claude mcp add -s user r-mcptools -- Rscript -e
"mcptools::mcp_server()"
```

3. In RStudio, to share your live session: `mcptools::mcp_session()`. To make it permanent, put `if (interactive()) mcptools::mcp_session()` in `.Rprofile`.
4. In Claude, type `/mcp` and look for **r-mcptools ✔ connected**.

4. Start a Project

1. Copy `part2_setup/project_template/` and rename it (lowercase, no spaces).
2. Fill in its `CLAUDE.md` (see `part2_setup/claude_md_guide.md`), and put your original files in `data_raw/`.
3. `cd` into the folder and run `claude`. For step-by-step prompts, copy in `part3_live_workflow/prompts/`.

```
# CLAUDE.md — [project]   (minimal; the template has 9
sections)
- Question: [1–2 sentences]. Data: [sources, years].
Tier: [de-identified / FERPA]. PHI: stop.
- Never modify data_raw/. Only 1000-series scripts
write to data_clean/.
- Scripts: code/NNNN_name.Rmd (0000 import · 1000 clean
· 2000 describe · 3000 model).
- Paths with here::here(). Never print individual
records; use counts.
- Never invent citations: write [CITATION NEEDED]. Ask
before research decisions.
```

5. Essential Commands

Command	What it does
`claude`	Start a session in this folder
`claude --continue`	Resume the last conversation here
`claude --resume`	Pick a past conversation
`/init`	Generate a starter `CLAUDE.md`
`/memory`	Edit the `CLAUDE.md` files in use
`/mcp`	Check connected tools
`/model`	Switch model

Command	What it does
`/compact`	Summarize to free up context
`/clear`	Start fresh in the same folder
`/usage`	Plan limits and usage
Shift+Tab	Cycle modes: Auto (default) · Manual · Accept edits · Plan
Esc / **Esc Esc**	Stop / rewind
Ctrl+L	Redraw a confusing screen

Learning? Use Manual mode (`claude --permission-mode default`). **Making a research-design decision? Use Plan mode**: Claude reads and proposes, but can't edit.

6. Two Engines

Gemini (free through WashU)	Claude CLI
Literature, long PDFs, many sources	Anything that touches your files
First-pass synthesis and framing	Writing and running `.Rmd` scripts
Drafting a spec or `CLAUDE.md` text	Cleaning, analysis, pipelines, drafts in `writing/`

Handoff: save Gemini's output as Markdown in `writing/`; Claude reads it from there.

7. Pocket Ethics Checklist

1. **Data tier:** public, de-identified, FERPA, PHI, or DUA-restricted?
2. **Workspace:** is it approved for that tier? **PHI never goes into Claude Code or any agentic tool.**
3. **Citations:** have I opened and verified every one?
4. **Disclosure:** did AI touch analysis, data, or results? Then say so in Methods: tool, version, date, what it did.
5. **Secrets:** no keys or passwords in `CLAUDE.md` or Git; keep them in `.Renviron`.

8. Resources

Free through WashU	Gemini for Education · Copilot Chat (Microsoft 365) · Ask WashU
Claude Edu	Free for eligible Danforth students (after the *AI Literacy* module); faculty and staff $16 or $50 per month
ChatGPT Edu	Faculty and staff, $10–$100 per month. Codex needs separate approval and is not approved for HIPAA or FERPA data

Personal Pro / Max	Works, but carries no WashU data agreement: keep university data out
Optional tools	**Warp** (free; has its own AI agent, so just type `claude`) · **Typora** ($14.99) · **Obsidian** (free)
Links	code.claude.com/docs · ai.washu.edu/tools

Prices and approvals checked 2026-10-12; they change often, so check the WashU pages.